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Cooperative game theory and cost allocation in energy communities

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Abstract

An energy community (EC) is a legal entity involving prosumers and consumers who produce, consume, and exchange energy. The members of these communities can cooperate to maximize the community’s social welfare. In practice, this naturally raises the question of cost sharing in the community, as the members may have different contributions to social welfare. In this paper, we empirically highlight the benefits of cooperation for the community and the individual members. Then, we present some cost-sharing mechanisms that guarantee fairness and the stability of the grand coalition composed of all prosumers and consumers. Finally, we present some results on instances built with real-world data from our partner Sween’s demonstrator, [Smart Lou Quila](#), in South France.

Keywords— Demand response, Storage, Electricity trading, Peer-to-peer exchanges, Stability, Fairness

1 Introduction

Renewable energies have been increasingly integrated into the energy mix and are expected to continue to do so in the future. This shift has implications for consumption patterns and the electricity grid. The first to emerge have been individual self-consumers, who generate energy for their own use, store some of it for later, and sell any excess on the energy market. This may allow them to reduce their energy bills. However, the growing number of self-consumers complicates the task of grid operators in maintaining a balance between electricity supply

and demand. Collective self-consumption communities (CSC) offer a potential solution to this issue. CSC is a virtual model where a group of end-users connected to the distribution network produce and consume energy Iazzolino et al. (2022). These communities can strive for self-sufficiency while also aligning with the [European Union](#)'s energy transition objectives. Several challenges need to be overcome to achieve the communities' objectives. These include maintaining a local balance between supply and demand and demand flexibility. On the one hand, this can be achieved through the local ownership of storage units that can help mitigate the effect of volatility in photovoltaic and wind generation. In addition, this balance can be ensured to some extent by shifting flexible loads to periods of renewable generation through demand response (DR) programs Bartolini et al. (2020); Sangaré et al. (2023).

This paper proposes a pricing model for CSCs connected to the distribution grid. In this model, prosumers can engage in peer-to-peer energy exchanges. The players are equipped with local generation technologies, e.g., solar PV panels, boilers, etc. They may additionally hold a storage unit. Each player can locally exchange her surplus, store it, or sell it to a leader. The leader is an end-user who collects the CSC's surpluses and supplies the excess demand if needed. The leader sets his exchange prices with the community to maximize his revenue.

There are various ways to implement DR in CSCs, with the most common method being price-based DR. This involves adjusting electricity prices for CSCs based on market prices over time. In the literature, authors often utilize bi-level optimization and game theory for price-based DR programs. In a particular model presented in Liu and Hu (2017), the authors consider renewable generation, even though community members do not personally own solar PV panels. Instead, renewable generation is viewed as a public resource. The authors propose a decentralized smart homes management model based on game theory, where an aggregator acts as a leader and sets differentiated discount factors for members. Since renewable generation is limited, community members engage in a noncooperative game to minimize their bills over time. In this paper, no price difference is allowed to accommodate [Smart Lou Quila \(SLQ\)](#)'s model, and the members can cooperate through peer-to-peer agreements. In the same line, Yu and Hong (2016) present a game theory model of the virtual interactions between an energy management center (EMC) and the electric devices of a smart home. In this model, EMC is the leader who sets a virtual energy selling price for each device to gain a virtual benefit, then each device optimizes its schedule to maximize its utility in response to the price signal. In our case, a member is not a device; it can be a residential, a stadium, an industrial end-user, etc, who owns a set of electrical devices to operate.

Regarding bi-level optimisation, note that for the sake of simplicity, in this paper, we use the pronoun “he” to refer to a leader and “she” to refer to a follower. That said, Mamounakis et al. (2019) present a model where an energy service provider (ESP) buys energy at time-varying prices from the wholesale market to sell it to a set of communities at his determined prices to maximize his profit. Then, the members of each community plan their loads to minimize their bills. Sarfarazi et al. (2023) present a bi-level model similar to the model in this paper. The authors consider a bi-level model where the leader is an aggregator who supplies/collects energy to/from a community and sets his interaction prices to maximize his revenue. There are several community members’ profiles related to asset ownership, and each member minimizes her bill. However, no peer-to-peer exchanges are not allowed.

In our paper, the leader determines the exchange prices within the community. These prices are not differentiated per member to maintain trust and comply with regulatory restrictions. Further, these prices depend on market prices and the community’s behavior. We are considering a centralized market design where a coordinator (CC) aims to maximize the overall benefit of the EC. After receiving the leader’s prices, CC plans the flexible demands (demand response), storage, peer-to-peer exchanges, and exchanges with the leader. Each community member sets their local exchange prices with other members. Therefore, the bi-level problem discussed in this paper puts a pricing problem at the upper level, while the coordinator’s problem is the lower-level problem; the price and the community’s surplus signals link both levels. Further, as the same leader price applies to all community members, we address the cost allocation to ensure each member is charged according to her contribution to social welfare. To do so, we model the community as a cooperative game and study the properties of this lower-level game.

Our contributions consist, on the one hand, of formulating a bilevel problem that simultaneously considers time-varying pricing, demand response, peer-to-peer exchanges, and energy storage units. Secondly, we have formulated and studied the properties of a cooperative game based on the outputs of the bilevel problem to ensure stability and fairness in the community. In addition, all theoretical results are illustrated using accurate data.

In the remainder of this paper, Section 1.1 presents the problem we are interested in. Section 2 presents the problem’s mathematical formulation and an analysis of the lower-level problem. Section 3 studies the cost allocation arising in the community. Finally, Sections 4 and 5 present some numerical results and our conclusions.

1.1 Description

We consider a set $\mathcal{N}$ of N end-users involved in an EC equipped with local PV solar panels or any other energy generation and storage unit. We also consider an agent, called *leader*, who interacts with the community to maximize his revenue. The members owning assets can exchange their surplus with other members or with the leader. They can also store this energy surplus in their storage unit if it exists. Each member n has a preference (the reservation price, for example) for the local peer-to-peer exchange with member m at period h denoted c_{nmh} . In addition, each member has a local demand to satisfy. Indeed, member n has a targeted non-flexible demand, the non-fulfillment of which results in a penalty, and a flexible demand which can be shifted over the planning horizon. Leader supplies the community when it is not self-sufficient and harvest the generation surplus when there is one; we assume the leader can collect all the community surpluses. The leader thus sets his periodic exchange prices with the community.

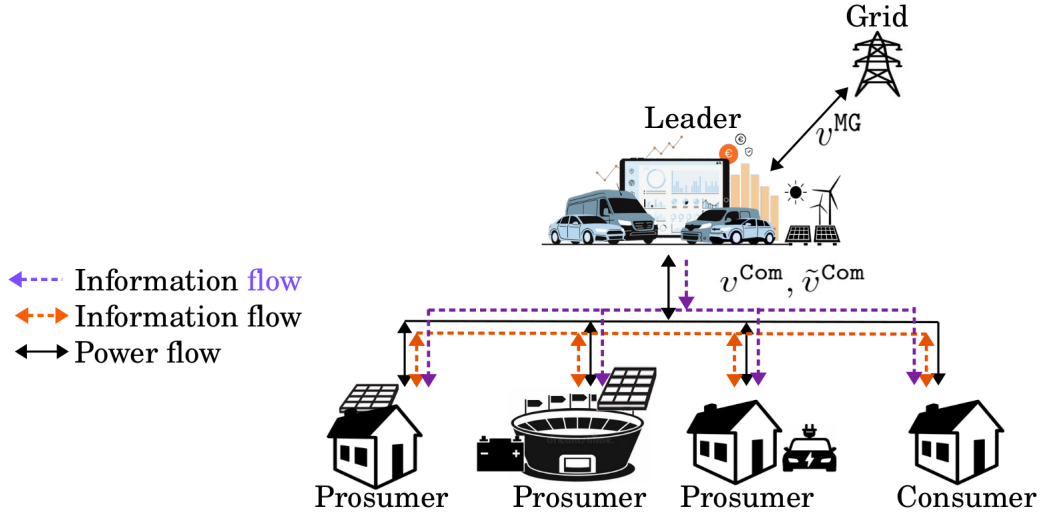


Figure 1: The market structure.

1.2 Market structure

In this paper, the market framework is presented in Figure 1 in the general form; it contains two types of participants: the leader and the end-users involved in an EC. An end-user in our community can be a prosumer or a consumer (see Figure 1). We discuss the participants' actions and roles later in the paper.

In this paper, the leader aims to maximize his revenue. We assume the leader owns a storage unit. At any given time, the leader ensures the balance between the community's offer and demand, supplying power to the community when there is not enough local generation and buying the generation surpluses when needed. The leader must set his energy prices to do so.

The other market participants are the community members, each needing energy to perform some tasks and may have some local generation or storage units. We assume that each community member owns an energy management system (EMS), allowing communication with the other market participants, computations, and other energy-related activities. The community members can perform peer-to-peer exchanges priced at a local exchange price. The overall market's problem can be formulated as a single leader multi-followers (members) bi-level problem formulated in (1).

$$\min_{x^L \in X^L(x^F), x^F \in X^F(x^L)} \Pi^L(x^L, x^F) \quad (1a)$$

$$\text{s.t.} \quad x^F \in \arg \max_{\tilde{x}^F \in X^F(x^L)} \Pi^F(x^L, \tilde{x}^F), \quad (1b)$$

where $X^L(x^F) = \{x^L \in \mathbb{R}^{n^1} : g^L(x^L, x^F) \leq 0\}$ and $X^F(x^L) = \{x^F \in \mathbb{R}^{n^2} : g^F(x^L, x^F) \leq 0\}$, $\Pi^L, \Pi^F : X^L(x^F) \times X^F(x^L) \rightarrow \mathbb{R}$, $g^L : X^L(x^F) \times X^F(x^L) \rightarrow \mathbb{R}^{\ell_1}$, and $g^F : X^L(x^F) \times X^F(x^L) \rightarrow \mathbb{R}^{\ell_2}$. Functions Π^L and g^L refer, respectively, to the leader's objective function and constraints, and (1b) to the follower's subproblem. Problem (1) is a single leader multi-followers problem. We detail it in the following sections.

2 Mathematical formulation

In this section, we formulate the bi-level problem. The problem has the following input data and variables, where the leader is indexed by "0".

Sets

- $\mathcal{H}$ set of periods in the planning horizon, we denote $|\mathcal{H}|$ as H .
- $\mathcal{N}$ set of members in the community, we denote $|\mathcal{N}|$ as N .
- $\mathcal{P}$ set of community members who can supply energy (the prosumers).
- $\mathcal{B}^{\text{BSS}}$ set of community members who own storage units $\mathcal{B}^{\text{BSS}} \subseteq \mathcal{P}$.

Parameters

- P_{nh}^{Gen} the amount of energy generated by prosumer $n \in \mathcal{P}$ at period h .

$\underline{d}_{nh}/\bar{d}_{nh}$	lower/upper-bound on n 's flexible demand at period h .
$\underline{D}_{nh}/\bar{D}_{nh}$	lower/upper-bound on member n 's non-flexible demand at period h .
D_{nh}^{Tar}	member n 's targeted non-flexible demand at period h .
s_{nh}	member n 's flexible demand at period h .
$\underline{\pi}_n/\bar{\pi}_n$	lower/upper bound on n 's subscribed capacity for $n \in \mathcal{N} \cup \{0\}$.
c_{nmh}	n 's preference for exchange with m at period h for $m \in \mathcal{P}, n \in \mathcal{N} : n \neq m$.
Γ_n	n 's storage capacity for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
ξ_n	n 's BSS's initial state of charge (SoC) for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
$\underline{E}_n/\bar{E}_n$	lower/upper-bound on n 's BSS's SoC for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
$\underline{Q}_n^1/\underline{Q}_n^2$	periodic minimum amount $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$ can charge/discharge into/from the BSS.
$\bar{Q}_n^1/\bar{Q}_n^2$	periodic maximum amount $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$ can charge/discharge into/from the BSS.
η_n	automatic discharge rate of n 's BSS for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
c_n/ℓ_n	n 's storage unit's charge/discharge efficiency for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
C_n^{BSS}	marginal cost of n 's BSS's operation for $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$.
$\underline{v}_h^b/\bar{v}_h^b$	leader's buying price's lower/upper bounds at period h .
$\underline{v}_h^s/\bar{v}_h^s$	leader's selling price's lower/upper bounds at period h .

Remark 1. Notice that parameters related to the storage units are null if a community member or the leader does not own the unit.

The leader's variables are the price v_h^{Com} at which she buys energy from the community and the price $\tilde{v}_h^{\text{Com}}$ at which she sells to the community. We associate to each community member n the decision variables D_{nh}^c and D_{nh} representing respectively her flexible and non-flexible demand at period h . Each BSS owner $n \in \mathcal{B}^{\text{BSS}} \cup \{0\}$ have variables q_{nh}^- and q_{nh}^+ which designate the amount of energy discharged from and charged into the BSS for $h \in \mathcal{H}$; variable E_{nh} represents the state of charge of these BSS. Finally, $f_{nmh} \in \mathbb{R}$ is the amount of energy exchanged between members n and $m \in \mathcal{P} \cup \{0\}$ (prosumer if $m \neq 0$ and the leader if $m = 0$) at period $h \in \mathcal{H}$. We adopt the following convention: member n sends energy to m if $f_{nmh} \geq 0$, and n receives energy from m if $f_{nmh} \leq 0$. The different levels are described in detail as follows.

2.1 Leader's problem: energy end-user (EEU)

We consider a situation where the leader is an agent who requires energy to carry out tasks. He can contract with an EC to buy or sell energy at attractive prices. This collaboration should benefit both parties as they exchange surplus for increased profitability. The leader determines the prices based on market rates. He may also own generation and storage units to meet his energy needs and satisfy the community's demand. Notice that if the leader cannot obtain sufficient energy from the community or his generation and storage units, he can collect energy from the main grid. The leader can capitalize on price fluctuations over time by owning a storage unit.

EEU can be seen as an entity that partners with rural communities isolated from the electrical grid for energy supplies. He can provide a public or private transportation service using an electric vehicle fleet, or more broadly, he can be a business located in EC's area that operates outside of the community. The leader's objective function is formulated as:

$$\Pi^L(x^L, x^F) = \sum_{h \in \mathcal{H}} \left[\sum_{n \in \mathcal{N}} (\tilde{v}_h^{\text{Com}} f_{0nh}^+ - v_h^{\text{Com}} f_{0nh}^-) - v_h^{\text{lead}} I_h - C_0^{\text{BSS}} (q_{0h}^- + q_{0h}^+) \right], \quad (2)$$

where variable $I_h \in \mathbb{R}$ is the amount of energy generated by the EEU at period h to satisfy his demand and supply the community (the amount I_h is locally generated or collected from the main grid); parameter v_h^{lead} is the related unitary generation cost.

In this paper, the leader's objective function depends on x^L and x^F . Specifically, EEU's profit contains the bilinear products between the followers' variables f_{0nh}^+ and f_{0nh}^- and the leader's prices $\tilde{v}_h^{\text{Com}}$ and v_h^{Com} for $n \in \mathcal{N}, h \in \mathcal{H}$. It also contains the leader's variables I_h related to EEU's generation and the storage cost related to variables q_{0h}^+ and q_{0h}^- for $h \in \mathcal{H}$. Indeed, function (2) combines the leader's revenue associated with his interactions with the community, the leader's generation, and the cost associated with the storage unit usage. The leader has a required demand d_h^{lead} at period h , which provides some positive revenue we do not consider. The leader's problem is:

$$\max_{x^L, x^F \in X^F} \Pi^L(x^L, x^F) \quad (3a)$$

$$v_h^s \leq \tilde{v}_h^{\text{Com}} \leq \bar{v}_h^s \quad h \in \mathcal{H} \quad (3b)$$

$$v_h^b \leq v_h^{\text{Com}} \leq \bar{v}_h^b \quad h \in \mathcal{H} \quad (3c)$$

$$\pi_0 \leq \sum_{n \in \mathcal{N}} f_{0nh} \leq \pi_0 \quad h \in \mathcal{H} \quad (3d)$$

$$\pi_0 \leq f_{0nh} \leq \pi_0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (3e)$$

$$\sum_{m \in \mathcal{P}} f_{mnh}^+ \perp \sum_{m \in \mathcal{P}} f_{mnh}^- \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (3f)$$

$$d_h^{\text{lead}} = I_h - \sum_{n \in \mathcal{N}} f_{0nh} + q_{0h}^- - q_{0h}^+ \quad h \in \mathcal{H} \quad (3g)$$

$$E_{0h} = \eta_0 E_{0(h-1)} + (c_0 q_{0h}^+ - d_0^{-1} q_{0h}^-)(\Gamma_0)^{-1} \quad h \in \mathcal{H} \quad (3h)$$

$$\underline{E}_0 \leq E_{0h} \leq \bar{E}_0 \quad h \in \mathcal{H} \quad (3i)$$

$$Q_0^1 \leq q_{0h}^+ \leq \bar{Q}_0^1 \quad h \in \mathcal{H} \quad (3j)$$

$$\underline{Q}_n^2 \leq q_{0h}^- \leq \bar{Q}_0^2 \quad h \in \mathcal{H} \quad (3k)$$

$$E_{00} = E_{0|H|} = \xi_0 \quad (3l)$$

$$I_h \geq 0 \quad h \in \mathcal{H} \quad (3m)$$

$$x^L \in \arg \max_{x^F \in X^F} \sum_{n \in \mathcal{N}} \Pi_n^F(x^L, x^F), \quad (3n)$$

where $f_{mnh} = f_{mnh}^+ - f_{mnh}^-$ and f_{mnh}^+ and f_{mnh}^- are non-negative, $x^L = (v_h^{\text{com}}, \tilde{v}_h^{\text{com}}, I_h, E_{0h}, q_{0h}^+, q_{0h}^-)_{h \in \mathcal{H}}$, x^F is the followers' decision variable, and $\Pi_n^F(x^L, x^F)$ is member n 's utility function which we describe in Section 2.2. Notice, the leader's feasible set X^L depends on the followers' variable x^F from constraints (3d)–(3f).

Constraints (3b)–(3c) set upper and lower bounds on the leader's prices, and constraints (3d)–(3e) ensure the leader's lines capacity are satisfied. These bounds are based on the prices set by the grid and ensure that the leader doesn't put his prices too high or too low for optimization purposes. Additionally, constraint (3f) prevents the leader from extracting and injecting energy simultaneously ($\perp$ means $\sum_{n \in \mathcal{P}} f_{mnh}^+$ and $\sum_{n \in \mathcal{P}} f_{mnh}^-$ cannot be simultaneously positive for member $n \in \mathcal{N}$ at period $h \in \mathcal{H}$). Lastly, the leader's interactions with the members depend on the community's objective value, depending on the peer-to-peer exchanges described in constraint (3n). Regarding the storage units, constraint (3g) is EEU's energy conservation constraint. Constraint (3h) computes EEU's BSS's SoC over the horizon and (3i)–(3l) set bounds an initial state on storage variables. We describe the community's problem in Section 2.2.

2.2 The followers' problem

The community coordinator is follower. The social welfare, which we denote $\Pi^F(x^L, x^F)$, corresponds to the sum of all community members' utility function $\sum_{n \in \mathcal{N}} \Pi_n^F(x^L, x^F)$. Each member decides her local consumption, storage, and exchanges and interacts with other members and the leader to contribute to the social welfare. The social welfare of the com-

munity takes the form:

$$\begin{aligned} \Pi^F(x^L, x^F) = \sum_{n \in \mathcal{N}} \Pi_n^F(x^L, x^F) = \sum_{n \in \mathcal{N}} \sum_{h \in \mathcal{H}} \left[U_{nh}(D_{nh}, D_{nh}^c) - G_{nh}^{\text{lead}}(f_{0nh}) \right. \\ \left. - G_{nh}^{\text{Com}} \left(\sum_{m \in \mathcal{P}: n \neq m} f_{mnh} \right) - G_{nh}^{\text{BSS}}(q_{nh}^+, q_{nh}^-) \right], \end{aligned} \quad (4)$$

where $\Pi_n^F(x^L, x^F)$ is the value of member $n \in \mathcal{N}$, and $G_{nh}^{\text{Com}}((f_{mnh})_{m \in \mathcal{P}: n \neq m})$, $G_{nh}^{\text{lead}}(f_{0nh})$ are the benefit of respectively exchange within the community and the leader, and $G_{nh}^{\text{BSS}}(q_{nh}^+, q_{nh}^-)$ is the cost associated to batteries' usage. Finally, function $U_{nh}(D_{nh}, D_{nh}^c)$ is member n 's satisfaction of having a certain demand level at period h .

Member n 's exchanges with the leader at period h yield to:

$$G_{nh}^{\text{lead}}(f_{0nh}) = \tilde{v}_h^{\text{Com}} f_{0nh}^+ - v_h^{\text{Com}} f_{0nh}^- \quad h \in \mathcal{H}, \quad (5)$$

which is an expense for $n \in \mathcal{N}$ if it is non-negative, a revenue otherwise. In the same line, the value associated with member n exchanges in the community is:

$$G_{nh}^{\text{Com}}((f_{mnh})_{m \in \mathcal{P}: n \neq m}) = \sum_{m \in \mathcal{P}: m \neq n} c_{nmh} f_{mnh} \quad h \in \mathcal{H}. \quad (6)$$

Member n 's battery usage at period h induces the cost G_{nh}^{BSS} defined in (7).

$$G_{nh}^{\text{BSS}}(q_{nh}^+, q_{nh}^-) = C_n^{\text{BSS}}(q_{nh}^+ + q_{nh}^-) \quad h \in \mathcal{H}, \quad (7)$$

recall that C_n^{BSS} is the marginal cost associated with member n 's battery usage.

The non-flexible loads have some periodic targeted demand, which induces some costs if they are unsatisfied. We model the load satisfaction as a concave function following Le Cadre et al. (2020). Thus, parameter $\tilde{a}_{nh} \geq 0$ is the penalty for the non-flexible loads' non-satisfaction. Each member sets her periodic targeted non-flexible demand D_{nh}^* and $\tilde{a}_{nh}$. The flexible loads are shiftable over the planning horizon. Each member sets the periodic targeted flexible load's demand s_{nh} and the periodic discomfort ℓ_{nh} caused by these consumption's shift. The utility associated with member n loads at period h is defined as:

$$U_{nh}(D_{nh}, D_{nh}^c) = -\tilde{a}_{nh}(D_{nh} - D_{nh}^{\text{Tar}})^2 - \ell_{nh}(D_{nh}^c - s_{nh})^2. \quad (8)$$

The first term in (8) is the uncontrollable load satisfaction, while the second refers to the flexible loads. As $\tilde{a}_{nh}$ and ℓ_{nh} are non-negative, and since a member n maximizes her welfare,

demand satisfactions are modeled using concave functions to incentivize the members to equalize the load with the periodic targeted demands. Otherwise, the member takes some penalties. Notice, the flexible loads satisfy constraint (9) for $n \in \mathcal{N}$:

$$\underline{d}_n = \min_{h \in \mathcal{H}} s_{nh} \leq D_{nh}^c \leq \max_{h \in \mathcal{H}} s_{nh} = \bar{d}_n, \quad (9)$$

which means the periodic flexible loads are between the minimum and the maximum level of the given targeted flexible demand s_{nh} for $n \in \mathcal{N}, h \in \mathcal{H}$ Liu et al. (2017).

Finally, the social welfare can be rewritten as:

$$\begin{aligned} \Pi^F(x^L, x^F) &= \sum_{n \in \mathcal{N}} \Pi_n^F(x^L, x^F) = \sum_{n \in \mathcal{N}} \sum_{h \in \mathcal{H}} [-\tilde{a}_{nh}(D_{nh} - D_{nh}^{\text{Tar}})^2 - \ell_{nh}(D_{nh}^c - s_{nh})^2 \\ &\quad - (\tilde{v}_h^{\text{Com}} f_{0nh}^+ - v_h^{\text{Com}} f_{0nh}^-) - \sum_{m \in \mathcal{P}: m \neq n} c_{nmh} f_{mnh} \\ &\quad - C_n^{\text{BSS}}(q_{nh}^+ + q_{nh}^-)]. \end{aligned} \quad (10)$$

In the remainder of this section, we assume we have a centralized market setting. In a centralized market setting, a community coordinator maximizes social welfare under community constraints. To do this, CC gathers information about each member, such as the generation capacity, storage unit characteristics, load profiles, and the subscribed capacity of prosumers. Thus, the coordinator solves (11) where the values in the bracket are the dual values associated with the corresponding constraints.

$$\max \quad \Pi^F((v_h^{\text{Com}}, \tilde{v}_h^{\text{Com}}, f_{mnh}, D_{nh}, D_{nh}^c, q_{nh}^+, q_{nh}^-, E_{nh})_{n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}}) \quad (11a)$$

$$\text{s.t.} \quad D_{nh} + D_{nh}^c = \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} f_{mnh} + P_{nh}^{\text{Gen}} + q_{nh}^- - q_{nh}^+ \quad n \in \mathcal{N}, h \in \mathcal{H} \quad [\alpha_{nh}] \quad (11b)$$

$$\underline{\pi}_n \leq \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} f_{mnh} \leq \bar{\pi}_n \quad n \in \mathcal{N}, h \in \mathcal{H} \quad [\gamma_{nh}, \bar{\gamma}_{nh}] \quad (11c)$$

$$\underline{\pi}_n \leq f_{mnh} \leq \bar{\pi}_n \quad n \in \mathcal{N}, m \in \mathcal{P} \cup \{0\}: m \neq n, h \in \mathcal{H} \quad [\theta_{mnh}, \bar{\theta}_{mnh}] \quad (11d)$$

$$f_{nmh} = -f_{mnh} \quad n \in \mathcal{N}, m \in \mathcal{P} \cup \{0\}: m \neq n, h \in \mathcal{H} \quad [\vartheta_{nmh}] \quad (11e)$$

$$\sum_{h \in \mathcal{H}} D_{nh}^c - s_{nh} = 0 \quad n \in \mathcal{N} \quad [o_n] \quad (11f)$$

$$E_{nh} = \eta_n E_{n(h-1)} + (c_n q_{nh}^+ - (d_n)^{-1} q_{nh}^-) \Gamma_n^{-1} \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad [\psi_{nh}] \quad (11g)$$

$$\underline{E}_n \leq E_{nh} \leq \bar{E}_n \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad [\mathcal{T}_{nh}, \bar{\mathcal{T}}_{nh}] \quad (11h)$$

$$\underline{Q}_n^1 \leq q_{nh}^+ \leq \bar{Q}_n^1 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad [\lambda_{nh}, \bar{\lambda}_{nh}] \quad (11i)$$

$$\underline{Q}_n^2 \leq q_{nh}^- \leq \bar{Q}_n^2 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad [\zeta_{nh}, \bar{\zeta}_{nh}] \quad (11j)$$

$$\underline{D}_{nh} \leq D_{nh} \leq \bar{D}_{nh} \quad n \in \mathcal{N}, h \in \mathcal{H} \quad [\varpi_{nh}, \bar{\varpi}_{nh}] \quad (11k)$$

$$\underline{d}_n \leq D_{nh}^c \leq \bar{d}_n \quad n \in \mathcal{N}, h \in \mathcal{H} \quad [\mu_{nh}, \bar{\mu}_{nh}] \quad (11l)$$

$$E_{n0} = E_{nH} = \xi_n \quad n \in \mathcal{B}^{\text{BSS}} \quad [\rho_n, \bar{\rho}_{nH}]. \quad (11m)$$

From (11), follower n 's feasibility set X_n^F does not depend on the leader's variable x^L . Thus, we refer to it as X_n^F in the following. Constraint (11b) represents the energy conservation condition. At each period h , member n 's demand is satisfied by the energy from the prosumers, the leader, the local generation, or the storage unit. Similarly, if member n has a generation surplus, it can be exchanged within the community, the leader, or stored in her local storage unit. Constraints (11c) and (11d) ensure the energy each member n sends and receives respect for the lines' capacity. Constraint (11e) imposes reciprocity between the sender and the receiver. Constraint (11f) ensures the total flexible loads' consumption must reach member n 's total required level over the horizon. Constraint (11g) allows computes the batteries' SoC and (11h) imposes the SoC into the specified bounds. The amounts injected into/out of batteries are limited by constraints (11i) and (11j). Constraints (11k) and (11l) ensure the demands are within their bounds. Finally, (11m) sets the batteries' initial and final SoC.

Each follower utility is a concave function to optimize over a convex set if we assume $\tilde{v}_h^{\text{Com}}$ to be possibly different from v_h^{Com} , $h \in \mathcal{H}$. The centralized community management solution is obtained using the Lagrangian function $\mathcal{L}(x^F)$:

$$\begin{aligned} \mathcal{L}(x^F) = & -\Pi^F((v_h^{\text{Com}}, \tilde{v}_h^{\text{Com}}, f_{mnh}, D_{nh}, D_{nh}^c, q_{nh}^+, q_{nh}^-, E_{nh})_{n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}}) \\ & + \sum_{n \in \mathcal{N}} \sum_{h \in \mathcal{H}} \left(\sum_{h \in \mathcal{H}} [\alpha_{nh}(D_{nh} + D_{nh}^c - \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} f_{mnh} - P_{nh}^{\text{Gen}} - q_{nh}^- + q_{nh}^+) \right. \\ & + \gamma_{nh}(\pi_n - \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} f_{mnh}) + \bar{\gamma}_{nh}(\sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} f_{mnh} - \bar{\pi}_n) \\ & + \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} \vartheta_{nmh}(f_{nmh} + f_{mnh}) + o_n(\sum_{h \in \mathcal{H}} D_{nh}^c - s_{nh}) \\ & + \sum_{m \in \mathcal{P} \cup \{0\}: m \neq n} \theta_{nmh}(\pi_n - f_{mnh}) + \bar{\theta}_{nmh}(f_{mnh} - \bar{\pi}_n) \\ & + \varpi_{nh}(\underline{D}_{nh} - D_{nh}) + \bar{\varpi}_{nh}(D_{nh} - \bar{D}_{nh}) + \mu_{nh}(\underline{d}_{nh} - D_{nh}^c) + \bar{\mu}_{nh}(D_{nh}^c - \bar{d}_{nh}) \Big) \\ & + \sum_{n \in \mathcal{B}^{\text{BSS}}} \left(\sum_{h \in \mathcal{H}} [\tau_{nh}(E_n - E_{nh}) + \bar{\tau}_{nh}(E_{nh} - \bar{E}_n) + \lambda_{nh}(Q_n^1 - q_{nh}^+) + \bar{\lambda}_{nh}(q_{nh}^+ - \bar{Q}_n^1) \right. \\ & \left. + \zeta_{nh}(Q_n^2 - q_{nh}^-) + \bar{\zeta}_{nh}(q_{nh}^- - \bar{Q}_n^2) \right] + \sigma_{n0}(E_{n0} - \xi_n) + \bar{\sigma}_{n|H|}(E_{n|H|} - \xi_n) \Big), \quad (12) \end{aligned}$$

where $x^F = (f_{mnh}, D_{nh}, D_{nh}^c, q_{nh}^+, q_{nh}^-, E_{nh}, \alpha_{nh}, \vartheta_{nmh}, \theta_{nmh}, \gamma_{nh}, \varpi_{nh}, \mu_{nh}, \sigma_{nh}, \tau_{nh}, \lambda_{nh}, \zeta_{nh}, o_n)_{n \in \mathcal{N}, m \in \mathcal{P}: n \neq m, h \in \mathcal{H}}$ is followers' variables. We compute the first order KKT conditions by deriving the Lagrangian function (12) associated to (11) with respect to $D_{nh}, D_{nh}^c, f_{0nh}^+, f_{0nh}^-, f_{mnh}, E_{nh}, q_{nh}^+$, and q_{nh}^- for $n \in \mathcal{N}, m \in \mathcal{P}$ such that $m \neq n, h \in \mathcal{H}$. Thus, the first order KKT conditions for $n \in \mathcal{N}, m \in \mathcal{P} : m \neq n, h \in \mathcal{H}$ are:

$$2\tilde{a}_{nh}(D_{nh} - D_{nh}^{\text{Tar}}) + \alpha_{nh} - \varpi_{nh} + \bar{\varpi}_{nh} = 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (13a)$$

$$2l_{nh}(D_{nh}^c - s_{nh}) + \alpha_{nh} - \underline{\mu}_{nh} + \bar{\mu}_{nh} + o_n = 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (13b)$$

$$C_n^{\text{BSS}} + \alpha_{nh} - \psi_{nh} c_n \Gamma_n^{-1} - \lambda_{nh} + \bar{\lambda}_{nh} = 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (13c)$$

$$C_n^{\text{BSS}} - \alpha_{nh} + \psi_{nh} (\mathcal{d})_n^{-1} \Gamma_n^{-1} - \zeta_{nh} + \bar{\zeta}_{nh} = 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (13d)$$

$$c_{nmh} + \alpha_{nh} - \vartheta_{nmh} - \underline{\gamma}_{nh} + \bar{\gamma}_{nh} - \underline{\theta}_{nmh} + \bar{\theta}_{nmh} = 0 \quad n \in \mathcal{N}, m \in \mathcal{P} : n \neq m, h \in \mathcal{H} \quad (13e)$$

$$\psi_{nh} - \eta_n \psi_{n(h+1)} - \underline{\tau}_{nh} + \bar{\tau}_{nh} = 0 \quad n \in \mathcal{N}, h \in \mathcal{H} : h < H \quad (13f)$$

$$-\eta_n \psi_{n0} + \sigma_{n0} = 0, \quad -\eta_n \psi_{nH} + \bar{\sigma}_{n|H|} = 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (13g)$$

$$\tilde{v}_h^{\text{Com}} - \alpha_{nh} - \underline{\gamma}_{nh} + \bar{\gamma}_{nh} - \underline{\theta}_{n0h} + \bar{\theta}_{n0h} = 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (13h)$$

$$v_h^{\text{Com}} - \alpha_{nh} - \underline{\gamma}_{nh} + \bar{\gamma}_{nh} - \underline{\theta}_{n0h} + \bar{\theta}_{n0h} = 0 \quad n \in \mathcal{N}, h \in \mathcal{H}. \quad (13i)$$

According to (13h) and (13i), $\tilde{v}_h^{\text{Com}} = v_h^{\text{Com}}$ for $h \in \mathcal{H}$. The complementary slackness conditions are:

$$0 \leq \underline{\gamma}_{nh} \perp \left(\sum_{m \in \mathcal{P} \cup \{0\}: n \neq m} f_{mnh} - \pi_n \right) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14a)$$

$$0 \leq \bar{\gamma}_{nh} \perp \left(\bar{\pi}_n - \sum_{m \in \mathcal{P} \cup \{0\}: n \neq m} f_{mnh} \right) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14b)$$

$$0 \leq \underline{\theta}_{nmh} \perp (f_{mnh} - \pi_n) \geq 0 \quad n \in \mathcal{N}, m \in \mathcal{P} \cup \{0\} : n \neq m, h \in \mathcal{H} \quad (14c)$$

$$0 \leq \bar{\theta}_{nmh} \perp (\bar{\pi}_n - f_{mnh}) \geq 0 \quad n \in \mathcal{N}, m \in \mathcal{P} \cup \{0\} : n \neq m, h \in \mathcal{H} \quad (14d)$$

$$0 \leq \varpi_{nh} \perp (D_{nh} - \underline{D}_{nh}) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14e)$$

$$0 \leq \bar{\varpi}_{nh} \perp (\bar{D}_{nh} - D_{nh}) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14f)$$

$$0 \leq \underline{\mu}_{nh} \perp (D_{nh}^c - \underline{d}_{nh}) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14g)$$

$$0 \leq \bar{\mu}_{nh} \perp (\bar{d}_{nh} - D_{nh}^c) \geq 0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (14h)$$

$$0 \leq \underline{\lambda}_{nh} \perp (q_{nh}^+ - Q_n^1) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (14i)$$

$$0 \leq \bar{\lambda}_{nh} \perp (\bar{Q}_n^1 - q_{nh}^+) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (14j)$$

$$0 \leq \underline{\zeta}_{nh} \perp (q_{nh}^- - Q_n^2) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (14k)$$

$$0 \leq \bar{\zeta}_{nh} \perp (\bar{Q}_n^2 - q_{nh}^-) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (14l)$$

$$0 \leq \mathcal{I}_{nh} \perp (E_{nh} - \underline{E}_n) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H} \quad (14\text{m})$$

$$0 \leq \bar{\tau}_{nh} \perp (\bar{E}_n - E_{nh}) \geq 0 \quad n \in \mathcal{B}^{\text{BSS}}, h \in \mathcal{H}. \quad (14\text{n})$$

Thus, the single-level reformulation of the problem is obtained by replacing constraint (3n) by the followers' problem's optimality conditions (13) and (14) and constraints (11b)–(11m). The single-level problem is:

$$\begin{aligned} \max_{x^L \in X^L(x^F), x^F \in X^F} \quad & \Pi^L(x^L, x^F) \\ \text{s.t.} \quad & (3\text{b}) - (3\text{n}), (11\text{b}) - (11\text{m}), (13), (14), \end{aligned} \quad (15\text{a})$$

which we can solve using the commercial solver as Gurobi under the non-convex mode for the complementarity constraints.

Further, given a solution $\tilde{x}^F$ of the lower-level problem, the leader's problem is feasible if $\tilde{x}^F$ satisfies constraints (16).

$$[\underline{v}_h^b, \bar{v}_h^b] \cap [\underline{v}_h^s, \bar{v}_h^s] \neq \emptyset \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (16\text{a})$$

$$c_{nmh} - v_h^{\text{com}} + 2\alpha_{nh} - \vartheta_{nmh} = 0 \quad n \in \mathcal{N}, m \in \mathcal{P} : n \neq m, h \in \mathcal{H} \quad (16\text{b})$$

$$\pi_0 \leq \sum_{n \in \mathcal{N}} f_{0nh} \leq \bar{\pi}_0 \quad n \in \mathcal{N}, h \in \mathcal{H} \quad (16\text{c})$$

$$\pi_0 \leq f_{0nh} \leq \bar{\pi}_0 \quad n \in \mathcal{N}, h \in \mathcal{H}. \quad (16\text{d})$$

Indeed, (16a) is required as $v_h^{\text{com}} = \tilde{v}_h^{\text{com}}$ for $h \in \mathcal{H}$ from (13h)–(13i). Condition (16b) comes from the optimality conditions (13e) and (13h). Finally, conditions (16c)–(16d) ensure that the leader's line capacity is respected.

2.3 Fairness issues

This section considers a coordinator who interacts with the leader and collects information from each community member to solve the problem (11). The resulting bilevel model, which we reformulate as (15), does not take into account any cost-sharing criteria and, further on, stability and fairness between members. However, the leader's prices depend on the members' aggregated behavior, raising some fairness issues as in Le Cadre et al. (2019). Thus, given the importance of these notions, in the following, we study the members' strategies considering fairness and stability criteria in a cooperative game setting.

We opt in what follows for ex-post cost allocation. Specifically, we assume that the prices of the leader are fixed and formulate the community problem for a given leader's price

$(v_h^{\text{Com}})_{h \in \mathcal{H}}$ as a cooperative game where n 's stand-alone utility is the optimal value of (11) by replacing $\mathcal{N}$ and $\mathcal{P}$ respectively by $\{n\}$ and $\{n, 0\}$.

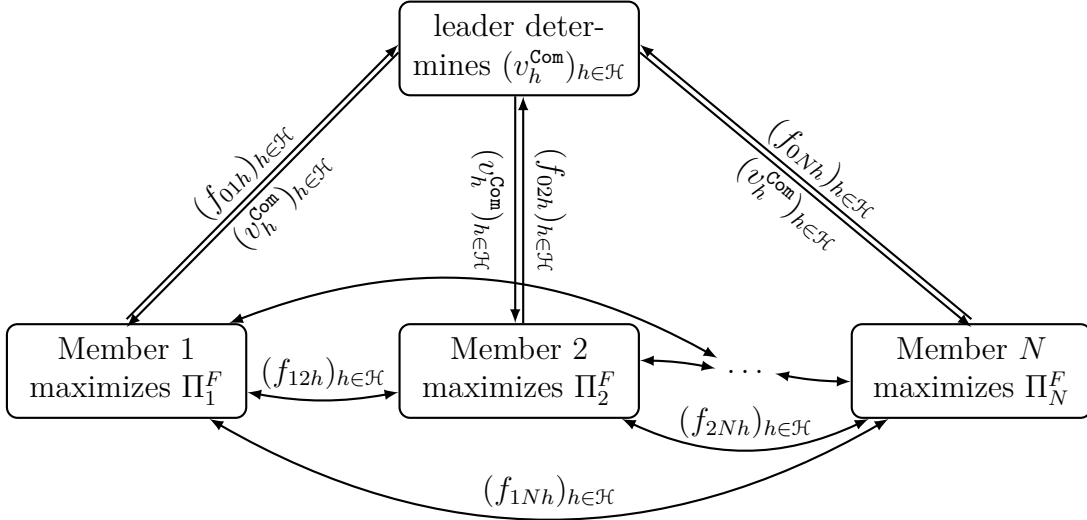


Figure 2: Interaction schemes.

3 Cost allocation

The previous lower-level formulation (11) may need more criteria like fairness and stability regarding cost sharing because the leader sets the same periodic price for each member. Thus, the price may not reflect their contribution to the supply-demand balance at the community scale. In this section, we formulate the community's problem as a cooperative game for $v_h^{\text{Com}} = \tilde{v}_h^{\text{Com}}$ fixed for each period $h \in \mathcal{H}$, and we study the different ways to share the social welfare to ensure fairness.

In the following, a *coalition* is a non-empty set of members who share cost according to a given mechanism. The set $\mathcal{N}$ of all members is called *grand coalition*.

Definition 1. A transferable utility game G is defined as a pair $(\mathcal{N}, \Psi)$, where $\mathcal{N}$ is the set of players, and $\Psi : 2^{\mathcal{N}} \rightarrow \mathbb{R}$ is the characteristic function which maps each coalition $\mathcal{S} \subseteq \mathcal{N}$ to a real value $\Psi(\mathcal{S})$ with $\Psi(\emptyset) = 0$.

Definition 2. Coalition structure Rahwan et al. (2015) Given a game $G = (\mathcal{N}, \Psi)$, a coalition structure over $\mathcal{N}$ is $CS = \{\mathcal{S}_1, \dots, \mathcal{S}_n\}$ a partition of $\mathcal{N}$ where $\mathcal{S}_i$ is a coalition, $i \in \{1, \dots, n\}$ i.e. $\cup_{i=1}^n \mathcal{S}_i = \mathcal{N}$ and $\mathcal{S}_i \cap \mathcal{S}_j = \emptyset$ for $i, j \in \{1, \dots, n\} : i \neq j$.

In the lower-level cooperative game, if a set of members agree to cooperate, they form a coalition in which members can exchange their surpluses.

Assumption 1. *Two disjoint coalitions do not interact, i.e., for $\mathcal{S}, \mathcal{S}' \subset \mathcal{CS}$,*

$$\mathcal{S} \cap \mathcal{S}' = \emptyset \Rightarrow f_{mnh} = 0 \quad n \in \mathcal{S}, m \in \mathcal{S}', h \in \mathcal{H}. \quad (17)$$

Recall that the leader does not intervene in the members' interactions. In fact, he just sets his interaction prices with the community; thus, no coalition can contain the leader.

In practice, it might be more attractive to form a single large community with respect to legislative and/or electrical constraints than to form multiple small renewable energy communities that do not interact. Thus, we assume that if the electrical and legislative constraints are satisfied, no community members should be incentivized to form a small community. If this assumption does not hold, the players can join any coalition. The coalition of all members is called the *grand coalition* in what follows, and each coalition $\mathcal{S} \subseteq \mathcal{N}$ value is defined as:

$$\begin{aligned} \Psi(\mathcal{S}) = & \sum_{n \in \mathcal{S}} \sum_{h \in \mathcal{H}} [-\tilde{a}_{nh}(D_{nh}^* - D_{nh}^{\text{Tar}})^2 - \ell_{nh}(D_{nh}^{\text{C}*} - s_{nh})^2 \\ & - \tilde{v}_h^{\text{Com}} f_{0nh}^* - \sum_{m \in \mathcal{P}: m \neq n} c_{nmh} f_{mnh}^* \\ & - C_n^{\text{BSS}}(q_{nh}^{+*} + q_{nh}^{-*})] \end{aligned} \quad \text{if } \mathcal{S} \neq \emptyset, 0 \text{ otherwise,} \quad (18)$$

where $(D_{nh}^*, D_{nh}^{\text{C}*}, f_{mnh}^*, q_{nh}^{+*}, q_{nh}^{-*})_{n \in \mathcal{N}, n \in \mathcal{P} \cap \mathcal{S}: n \neq m, h \in \mathcal{H}}$ is related respectively to the members' optimal demands, their optimal interactions and storage units usage with respect to (11). Thus, the cooperative game which arises in the community denoted G is the pair $(\mathcal{N}, \Psi)$, where $\mathcal{N}$ is the set of all players (the community members), and function $\Psi : 2^{\mathcal{N}} \rightarrow \mathbb{R}^{\mathcal{N}}$ assigns a value to each non-empty coalition of players as defined in (18).

In the remainder of this paper, we consider the following definitions:

Definition 3. *Value $\tilde{\Pi}_n$ is the optimal solution of problem (11) if $\mathcal{N} = \{n\}$ and $\mathcal{P} = \{n, 0\}$ i.e. $\tilde{\Pi}_n$ is member n 's optimal value while not cooperating with other community members.*

Definition 4 (Cost allocation Sanjab et al. (2022)). *A vector $y(G) = \{y_1(G), \dots, y_N(G)\}$ is a cost allocation for coalition structure $\mathcal{CS} = \{\mathcal{S}_1, \dots, \mathcal{S}_n\}$ over $\mathcal{N}$ if $\sum_{n \in \mathcal{S}_i} y_n(G) \geq \Psi(\mathcal{S}_i)$ for $i \in \{1, \dots, n\}$.*

Definition 5 (Efficiency Tijs and Driessen (1986)). *A cost allocation $y(G)$ is efficient if value $\Psi(\mathcal{S}_i)$ of coalition $\mathcal{S}_i$ is totally distributed among the members of the coalition i.e. if $\sum_{n \in \mathcal{S}_i} y_n(G) = \Psi(\mathcal{S}_i)$ for $i \in \{1, \dots, n\}$.*

Definition 6 (Individually rationality Branzei et al. (2008)). *A cost allocation mechanism $y(G)$ is considered individually rational if no member receives less than their value when operating alone, i.e., if $y_n(G) \geq \tilde{\Pi}_n^F$ for each $n \in \mathcal{N}$.*

A set $\mathcal{S} \subseteq \mathcal{N}$ of members have the incentive to leave the grand coalition $\mathcal{N}$ if $\sum_{n \in \mathcal{S}} y_n(G) < \Psi(\mathcal{S})$. The set of cost allocations, such that no member has incentives to leave the grand coalition, is called the core of G .

Definition 7 (Core Shapley (1971)). *The core of a game G contains the allocation mechanisms y that satisfy:*

$$\mathcal{C} = \left\{ y(G) \in \mathbb{R}^N : \sum_{n \in \mathcal{S}} y_n(G) \geq \Psi(\mathcal{S}) \text{ for } \mathcal{S} \subset \mathcal{N} \text{ and } \sum_{m \in \mathcal{N}} y_m(G) = \Psi(\mathcal{N}) \right\}. \quad (19)$$

In other words, if it is not empty, the core of a game contains the mechanisms that satisfy efficiency and individual rationality Tijs and Driessen (1986).

From Shapley (1967), a collection $\mathcal{B}$ of coalitions is said to be balanced if there exist positive numbers $\varsigma_{\mathcal{S}}$ for $\mathcal{S} \in \mathcal{B}$ such that for $n \in \mathcal{S}$,

$$\sum_{\mathcal{S} \in \mathcal{B}, n \in \mathcal{S}} \varsigma_{\mathcal{S}} = 1.$$

A game $G = (\mathcal{N}, \Psi)$ is balanced Shapley (1967) if, for each balanced collection $\mathcal{B}$ with balancing weight $\varsigma_{\mathcal{S}}$:

$$\sum_{\mathcal{S} \in \mathcal{B}} \varsigma_{\mathcal{S}} \Psi(\mathcal{S}) \leq \Psi(\mathcal{N}).$$

The author shows the non-emptiness of balanced games' core.

Definition 8 (Superadditivity Shapley (1971)). *A game $G = (\mathcal{N}, \varrho)$ is called superadditive if $\varrho(\mathcal{S} \cup \mathcal{S}') \geq \varrho(\mathcal{S}) + \varrho(\mathcal{S}')$ for any two disjoint coalitions $\mathcal{S}$ and $\mathcal{S}'$.*

Definition 9 (Supermodularity Shapley (1971)). *A characteristic function ϱ is supermodular if for each $n \in \mathcal{N}, \mathcal{S} \subseteq \mathcal{S}' \subseteq \mathcal{N} \setminus \{n\}$,*

$$\varrho(\mathcal{S} \cup \{n\}) - \varrho(\mathcal{S}) \leq \varrho(\mathcal{S}' \cup \{n\}) - \varrho(\mathcal{S}'). \quad (20)$$

Proposition 1. *Game $G = (\mathcal{N}, \Psi)$ is superadditive.*

Proof. As Ψ results from a maximization problem, joining two disjoint communities increases the set of optimal interactions as all decisions available in the disjoint set are available in the joined set. $\square$

Proposition 2. *The core of game $G = (\mathcal{N}, \Psi)$ is non-empty.*

Proof. From Shapley (1967), $\mathcal{C}$ is nonempty if game G is balanced. The set of balanced games contains the games that derive from an exchange economy in which the traders have continuous concave utility functions Shapley and Shubik (1969). Thus, since each member optimizes a continuous and concave utility function under linear constraints, $\mathcal{C}$ is not empty. $\square$

The core refers to the allocation mechanisms where no member has an incentive to leave the grand coalition. In Section 3.1, we examine the properties of various mechanisms (from literature) for the game G .

Proposition 3. *If $c_{nmh} = c_{mnh}$ for $n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}$, $\Psi(\mathcal{S}) = \sum_{n \in \mathcal{S}} \tilde{\Pi}_n^F$ for $\mathcal{S} \subseteq \mathcal{N}$.*

Proof. Let us recall that $\tilde{\Pi}_n^F$ is the optimal solution of problem (11) replacing respectively $\mathcal{N}$ and $\mathcal{P}$ by $\{n\}$ and $\{n, 0\}$. Also, $\Psi(\mathcal{S})$ is defined as the optimal solution cost of problem (11) considering the sets $\mathcal{S}$ and $\mathcal{S} \cap \mathcal{P}$ instead of $\mathcal{N}$ and $\mathcal{P}$.

Since $f_{nmh} = -f_{mnh}$ from constraint (11e) and $c_{nmh} = c_{mnh}$ for $n \in \mathcal{S}, m \in \mathcal{S} \cap \mathcal{P} : n \neq m, h \in \mathcal{H}$, the objective function $\Pi_{\mathcal{S}}^F(\tilde{x}^L, x^F)$ of coalition $\mathcal{S}$ takes the form:

$$\Pi_{\mathcal{S}}^F(\tilde{x}^L, x^F) = \sum_{n \in \mathcal{S}} \sum_{h \in \mathcal{H}} [U_{nh}(D_{nh}, D_{nh}^C) - v_h^{\text{Com}} f_{0nh} - C_n^{\text{BSS}}(q_{nh}^+ + q_{nh}^-)].$$

Given an optimal solution $\tilde{x}^F = (\tilde{D}_{nh}, \tilde{D}_{nh}^C, \tilde{q}_{nh}^-, \tilde{q}_{nh}^+, (\tilde{f}_{mnh})_{m \in \mathcal{S} \cap \mathcal{P} : n \neq m}, \tilde{f}_{0nh})_{n \in \mathcal{S}, h \in \mathcal{H}}$ of coalition $\mathcal{S}$'s problem, $\tilde{x}^F$ is equivalent to $\tilde{\tilde{x}}^F = (\tilde{D}_{nh}, \tilde{D}_{nh}^C, \tilde{q}_{nh}^-, \tilde{q}_{nh}^+, \tilde{\tilde{f}}_{0nh})_{n \in \mathcal{S}, h \in \mathcal{H}}$ where $\tilde{\tilde{f}}_{0nh} = \tilde{f}_{0nh} + \sum_{m \in \mathcal{S} \cap \mathcal{P} : n \neq m} \tilde{f}_{mnh}$ for $n \in \mathcal{S}, h \in \mathcal{H}$. In other words, $\tilde{\tilde{x}}^F$ is feasible and does not change the objective value $\Pi_{\mathcal{S}}^F(\tilde{x}^L, \tilde{\tilde{x}}^F)$. In fact, the leader can take 100% of the community surpluses.

In addition,

$$\begin{aligned}
\Pi_S^F(\tilde{x}^L, \tilde{x}^F) &= \sum_{n \in \mathcal{S}} \sum_{h \in \mathcal{H}} [U_{nh}(\tilde{D}_{nh}, \tilde{D}_{nh}^C) - v_h^{\text{com}} \tilde{f}_{0nh} - C_n^{\text{BSS}}(\tilde{q}_{nh}^+ + \tilde{q}_{nh}^-)] \\
&= \sum_{n \in \mathcal{S}} \sum_{h \in \mathcal{H}} [U_{nh}(\tilde{D}_{nh}, \tilde{D}_{nh}^C) - v_h^{\text{com}} (\tilde{f}_{0nh} + \sum_{m \in \mathcal{S} \cap \mathcal{P}} \tilde{f}_{mnh}) - C_n^{\text{BSS}}(\tilde{q}_{nh}^+ + \tilde{q}_{nh}^-)] \\
&= \sum_{n \in \mathcal{S}} \left[\sum_{h \in \mathcal{H}} U_{nh}(\tilde{D}_{nh}, \tilde{D}_{nh}^C) - v_h^{\text{com}} \tilde{f}_{0nh} - C_n^{\text{BSS}}(\tilde{q}_{nh}^+ + \tilde{q}_{nh}^-) \right] \\
&\quad - \sum_{h \in \mathcal{H}} v_h^{\text{com}} \sum_{n \in \mathcal{S}} \sum_{m \in \mathcal{S} \cap \mathcal{P}: n \neq m} \tilde{f}_{mnh} \\
&= \Pi_S^F(\tilde{x}^L, \tilde{x}^F) - \sum_{h \in \mathcal{H}} v_h^{\text{com}} \sum_{n \in \mathcal{S}} \sum_{m \in \mathcal{S} \cap \mathcal{P}: n \neq m} \tilde{f}_{mnh}.
\end{aligned}$$

Since $\tilde{x}^F$ is a solution for coalition $\mathcal{S}$ problem, $\tilde{f}_{mnh} + \tilde{f}_{nmh} = 0$ for $n \in \mathcal{S}, m \in \mathcal{S} \cap \mathcal{P} : n \neq m, h \in \mathcal{H}$. As a consequence, $\sum_{n \in \mathcal{S}} \sum_{m \in \mathcal{S} \cap \mathcal{P}: n \neq m} \tilde{f}_{mnh} = 0$ for $h \in \mathcal{H}$. Therefore, at optimality:

$$\Psi(\mathcal{S}) = \Pi_S^F(\tilde{x}^L, \tilde{x}^F) = \Pi_S^F(\tilde{x}^L, \tilde{x}^F) = \sum_{n \in \mathcal{S}} \tilde{\Pi}_n^F \quad \mathcal{S} \subseteq \mathcal{N}.$$

□

Corollary 1. *The characteristic function Ψ is supermodular if $c_{nmh} = c_{mnh}$ for $n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}$.*

Proof. From Proposition 3, for each $n \in \mathcal{N}, \mathcal{S} \subseteq \mathcal{S}' \subseteq \mathcal{N} \setminus \{n\}$, we have:

$$\begin{aligned}
\Psi(\mathcal{S} \cup \{n\}) - \Psi(\mathcal{S}) &= \sum_{m \in \mathcal{S} \cup \{n\}} \tilde{\Pi}_m^F - \sum_{m \in \mathcal{S}} \tilde{\Pi}_m^F = \tilde{\Pi}_n^F \\
\Psi(\mathcal{S}' \cup \{n\}) - \Psi(\mathcal{S}') &= \sum_{m \in \mathcal{S}' \cup \{n\}} \tilde{\Pi}_m^F - \sum_{m \in \mathcal{S}'} \tilde{\Pi}_m^F = \tilde{\Pi}_n^F
\end{aligned}$$

According to Definition 9, Ψ is supermodular. As a consequence, from Shapley (1971) game $G = (\mathcal{N}, \Psi)$ is convex. □

Now that we have formulated the lower-level problem as a cooperative game and examined its properties, Section 3.1 delves into the properties of various cost allocation mechanisms of the literature for game G .

3.1 Allocation mechanisms

As the game $G = (\mathcal{N}, \Psi)$ is stable, we introduce and evaluate some cost allocation mechanisms of the literature and study them according to a set of properties. A cost allocation mechanism $\mathcal{Y}(G)$ in this paper is evaluated according to the following properties defined in Chalkiadakis et al. (2011):

- (i) **efficiency**: $\mathcal{Y}(G)$ is efficient if $\sum_{n \in \mathcal{N}} \mathcal{Y}_n(G) = \Psi(\mathcal{N})$.
- (ii) **dummy member**: if $\Psi(\mathcal{S} \cup \{n\}) = \Psi(\mathcal{S}) \forall \mathcal{S} \subseteq \mathcal{N} \setminus \{n\}$, then $\mathcal{Y}_n(G) = 0$.
- (iii) **additivity**: if Ψ decomposes into Ψ' and Ψ'' , $\mathcal{Y}_n(G) = \mathcal{Y}_n((\mathcal{N}, \Psi')) + \mathcal{Y}_n((\mathcal{N}, \Psi''))$, $\forall n \in \mathcal{N}$.
- (iv) **symmetry**: if $\Psi(\mathcal{S} \cup \{n\}) = \Psi(\mathcal{S} \cup \{n'\}) \forall \mathcal{S} \subseteq \mathcal{N}$, then $\mathcal{Y}_n(G) = \mathcal{Y}_{n'}(G)$.
- (v) **anonymity**: for two players n and n' , if $\tilde{G}$ is a game identical to G except the roles of players n and n' are exchanged, then $\mathcal{Y}_n(G) = \mathcal{Y}_{n'}(\tilde{G})$.
- (vi) **stability**: the cost allocation $\mathcal{Y}(G)$ is said to be stable if it belongs to the core (19).

Table 1 presents the mechanism properties in game G . We list next some allocation mechanisms.

Equal sharing (EQ). This allocation mechanism shares the cost equally among the members. Thus, let us denote w_n as member n 's weight, $w_n = N^{-1}$, each member receives $y_n(G) = w_n \Psi(\mathcal{N})$.

The equal sharing mechanism satisfies (i), (iii), and (iv). It is not dummy since a member who does not generate any cost is allocated. It satisfies (vi) if:

$$S \cdot N^{-1} \leq \Psi(\mathcal{S}) \Psi(\mathcal{N})^{-1} \quad \mathcal{S} \subset \mathcal{N}, \quad (21)$$

where $S = |\mathcal{S}|$. The equal sharing mechanism meets (v).

Normalized Banzhaf index ($\mathcal{B}^\#$) Chalkiadakis et al. (2011). The Banzhaf index measures the a priori power of each player of a game. Member n 's Banzhaf index denoted $\mathcal{B}_n(G)$ is:

$$\mathcal{B}_n(G) = \frac{1}{2^{N-1}} \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} [\Psi(\mathcal{S} \cup \{n\}) - \Psi(\mathcal{S})]. \quad (22)$$

The Banzhaf value measures the expected marginal contribution of member n and the average overall coalitions of the game. It satisfies (ii)–(iv). It does not satisfy (i). Thus, the normalized Banzhaf index (23), which satisfies efficiency, is considered.

$$\mathcal{B}_n^\#(G) = \Psi(\mathcal{N}) \cdot \mathcal{B}_n(G) \cdot \left(\sum_{m \in \mathcal{N}} \mathcal{B}_m(G) \right)^{-1}. \quad (23)$$

While this version of the Banzhaf index satisfies efficiency, it loses the additivity. It satisfies (v). When it comes to (vi), the normalized Banzhaf index is a convex combination of elements constructed in the proof of Th.2.27 in Chalkiadakis et al. (2011) to exhibit elements from the core of convex games. Since these elements belong to the core by construction, the normalized Banzhaf index is a convex combination of these elements, and the core of game G is a convex set. Hence, the normalized Banzhaf index meets individual rationality. As a consequence, the normalized Banzhaf index meets (vi).

Proportional cost allocation (PCA) Tijs and Driessen (1986). This mechanism shares the cost $\Psi(\mathcal{N})$ of the grand coalition among the members according to the volume of flexibility offered/used by each of them Sanjab et al. (2022). Thus:

$$\text{PCA}_n(G) = w_n \Psi(\mathcal{N}), \quad (24)$$

where member n 's weight is: $w_n = \tilde{\Pi}_n^F (\sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F)^{-1}$. Recall $\tilde{\Pi}_n^F$ is member n 's value while not cooperating with any community member (Definition 3).

PCA satisfies (i)–(iv), and (v). Furthermore, it satisfies (vi) under condition (25).

$$\sum_{n \in \mathcal{S}} w_n \geq \Psi(\mathcal{S}) \Psi(\mathcal{N})^{-1} \quad \mathcal{S} \subset \mathcal{N}. \quad (25)$$

The Shapley value (SV) Shapley (1953). The Shapley value allocates to members according to their contribution to all (sub) coalitions. The value allocated to n is:

$$SV_n(G) = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} [\Psi(\mathcal{S} \cup \{n\}) - \Psi(\mathcal{S})]. \quad (26)$$

where $w_{\mathcal{S}} = S!(N - S - 1)!(N!)^{-1}$. The Shapley value satisfies (i)–(iv) from Shapley (1953), it also meets (v). From Shapley (1971), since $G = (\mathcal{N}, \Psi)$ is convex, the Shapley value satisfies (vi).

Equal profit method (EPM) Frisk et al. (2010). The equal profit method is a stable sharing method that minimizes the maximum difference between each pair of members. This allocation is the solution y of the linear program (27).

$$\min \quad f \quad (27a)$$

$$\text{s.t.} \quad f \geq \frac{\text{EPM}_n(G)}{\tilde{\Pi}_n^F} - \frac{\text{EPM}_m(G)}{\tilde{\Pi}_m^F} \quad n, m \in \mathcal{N} : n \neq m \quad (27b)$$

$$\sum_{n \in \mathcal{S}} \text{EPM}_n(G) \geq \Psi(\mathcal{S}) \quad \mathcal{S} \subset \mathcal{N} \quad (27c)$$

$$\sum_{n \in \mathcal{N}} \text{EPM}_n(G) = \Psi(\mathcal{N}). \quad (27d)$$

It satisfies (iv) by constraint (27b) and (i) by (27d). EPM belongs to the game's core by construction due to constraints (27c)–(27d). The mechanism does not satisfy (v). However, as members' values can be positive or negative, constraint (27b) can create infeasibilities in model (27). In other words, EPM is not well defined.

Proposition 4. *In the grand coalition, if $c_{nmh} = c_{mnh}$ for $n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}$, the mechanisms SV, $\mathcal{B}^\#$, PCA and EPM give the same cost allocation values.*

Proof. Herein, we leverage Proposition 3 to show that the mechanisms allocate identically. Indeed, as $c_{nmh} = c_{mnh}$ for $n \in \mathcal{N}, m \in \mathcal{P}, h \in \mathcal{H}$, we have:

$$\Psi(\mathcal{S}) = \sum_{n \in \mathcal{S}} \tilde{\Pi}_n^F \quad \text{for } \mathcal{S} \subseteq \mathcal{N}.$$

Regarding the Shapley value (26), for each member $n \in \mathcal{N}$, we have:

$$\begin{aligned} \text{SV}_n(G) &= \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} [\Psi(\mathcal{S} \cup \{n\}) - \Psi(\mathcal{S})] = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} \left[\sum_{m \in \mathcal{S} \cup \{n\}} \tilde{\Pi}_m^F - \sum_{m \in \mathcal{S}} \tilde{\Pi}_m^F \right] \\ &= \tilde{\Pi}_n^F \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}}, \end{aligned} \quad (28)$$

where $w_{\mathcal{S}} = (S!(N - S - 1!)(N!)^{-1}$ for $\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}$ and

$$\sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} (S!(N - S - 1!)(N!)^{-1} = 1.$$

By replacing $\sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}}$ by 1 in (28), we obtain $\text{SV}_n(G) = \tilde{\Pi}_n^F$.

Coming to the Banzhaf index, for $n \in \mathcal{N}$:

$$\mathcal{B}_n(G) = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} [\Psi(\mathcal{S} \cup \{n\}) - \Psi(\mathcal{S})] = \tilde{\Pi}_n^F \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}}, \quad (29)$$

where $w_{\mathcal{S}} = 2^{1-N}$ for $\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}$ and

$$\sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}} = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} 2^{1-N} = 1,$$

By replacing $\sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{n\}} w_{\mathcal{S}}$ by 1 in (29), we obtain $\mathcal{B}_n(G) = \tilde{\Pi}_n^F$. As a consequence, the normalized Banzhaf index $\mathcal{B}_n^{\#}(G)$ for $n \in \mathcal{N}$ defined in (23) becomes:

$$\mathcal{B}_n^{\#}(G) = \left(\sum_{n \in \mathcal{N}} \tilde{\Pi}_n^F \right) \tilde{\Pi}_n^F \left(\sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F \right)^{-1} = \tilde{\Pi}_n^F.$$

Further, $\text{PCA}_n(G) = w_n \Psi(\mathcal{N})$ where $w_n = \tilde{\Pi}_n^F (\sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F)^{-1}$. Then,

$$\text{PCA}_n(G) = \tilde{\Pi}_n^F \left(\sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F \right)^{-1} \Psi(\mathcal{N}) = \tilde{\Pi}_n^F \left(\sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F \right)^{-1} \sum_{m \in \mathcal{N}} \tilde{\Pi}_m^F = \tilde{\Pi}_n^F.$$

Finally, regarding EPM, the efficiency constraint (27d) implies:

$$\sum_{n \in \mathcal{N}} \text{EPM}_n(G) = \Psi(\mathcal{N}) = \sum_{n \in \mathcal{N}} \tilde{\Pi}_n^F \Leftrightarrow \sum_{n \in \mathcal{N}} (\text{EPM}_n(G) - \tilde{\Pi}_n^F) = 0.$$

$\text{EPM}_n(G)$ satisfies constraint (27c), which implies $\text{EPM}_n(G) \geq \tilde{\Pi}_n^F$ for $n \in \mathcal{N}$, efficiency $\sum_{n \in \mathcal{N}} (\text{EPM}_n(G) - \tilde{\Pi}_n^F) = 0$ is equivalent to $\text{EPM}(G)_n = \tilde{\Pi}_n^F$ for $n \in \mathcal{N}$. $\square$

Property	EQ(G)	$\mathcal{B}^{\#}(G)$	SV(G)	PCA(G)	EPM(G)
(i)	✓	✓	✓	✓	✓
(ii)	✗	✓	✓	✗	✗
(iii)	✓	✗	✓	✓	✗
(iv)	✓	✓	✓	✓	✓
(v)	✓	✓	✓	✗	✗
(vi)	✓ under (21)	✓	✓	✓ under (25)	✓ if (27) is feasible

Table 1: Cost allocation methods' properties for cost game $G = (\mathcal{N}, \Psi)$.

4 Numerical experiments

This section provides numerical results based on realistic instances using data from our partner’s demonstrator, Smart Lou Quila. Our instances are detailed in Section 4.1, while the results are presented in Section 4.2. In the latter section, for each instance, we solve the single-level problem (15) with a specified leader’s objective to determine the leader’s price and the grand coalition’s value. Subsequently, we share the grand coalition’s value among the members according to the allocation mechanisms presented in Section 3.1.

4.1 Instances’ description

4.1.1 Leader

We set the leader’s prices bounds $\underline{v}_h^s = \underline{v}_h^b = 0.1$ and $\bar{v}_h^s = \bar{v}_h^b = 0.9$ for $h \in \mathcal{H}$, the leader’s unitary cost $v_h^{\text{lead}} = 0.91$ for $h \in \mathcal{H}$. In addition, the leader’s subscribed capacity is assumed to be $\pi_0 = 4N$ and $\underline{\pi}_0 = -4N$. These values are the same no matter the leader’s objective. While being an EEU, the leader needs five kWh at each period for simplicity; $d_h^{\text{lead}} = 5$ for $h \in \mathcal{H}$. She owns a BSS, that has a capacity of $\Gamma_0 = 32$ kWh, the minimum respectively maximum energy that can be charged/discharged per period into/from it are $\underline{Q}_0^1 = \underline{Q}_0^2 = 0$ and $\bar{Q}_0^1 = \bar{Q}_0^2 = 16$. The BSS’s charge/discharge efficiency is $c_0 = d_0 = 0.98$, and the automatic discharge rate is $\eta_0 = 1\%$. Finally, the unitary cost associated with the BSS’s usage is $C_0^{\text{BSS}} = 0.1$.

4.1.2 Community

We consider a community of at most ten members; we use some real-world data as members’ generation and loads. Specifically, we use the net generations and injections of the members in [Smart Lou Quila](#). These data are collected for May 2023, and members 1, 4, 5, and 10 have some generation over the horizon, and 2, 3, 5, 6, 7, 8, and 9 do not generate energy. The planning is performed on a day sliced into 12 periods for each instance, i.e., $H = 12$. Regarding the non-flexible loads, Figure 3b presents D_{nh}^{Tar} for $n \in \mathcal{N}, h \in \mathcal{H}$. We set the penalty related to non-flexible loads $\tilde{a}_{nh} = n$ for $n \in \mathcal{N}, h \in \mathcal{H}$. Further, we assume that members 2 and 3 have some flexible loads. Figure 3c shows this flexible demand over time; we set the penalty related to these loads $\ell_{nh} = 0.5$ for $n \in \{2, 3\}, h \in \mathcal{H}$.

Each member’s subscribed capacity is set to 4; $\pi_n = 4$ and $\underline{\pi}_n = -4$ for $n \in \mathcal{N}$. Figure 3 presents the members’ load profiles for each instance. We consider different BSS owing situations. If a member owns a BSS, it has a capacity of $\Gamma = 16$ kWh, the minimum

respectively maximum energy that can be charged/discharged per period into/from it are $\underline{Q}^1 = \underline{Q}^2 = 0$ and $\bar{Q}^1 = \bar{Q}^2 = 8$. The BSS's charge/discharge efficiency is $c = d = 0.98$, and the automatic discharge rate is $\eta = 1\%$. Finally, the unitary cost associated with the BSS's usage is $C^{\text{BSS}} = 0.2$. The data and model presented in this paper are available at <https://github.com/MsangL/EnergyCommunitiesBilevelOptimization.git>.

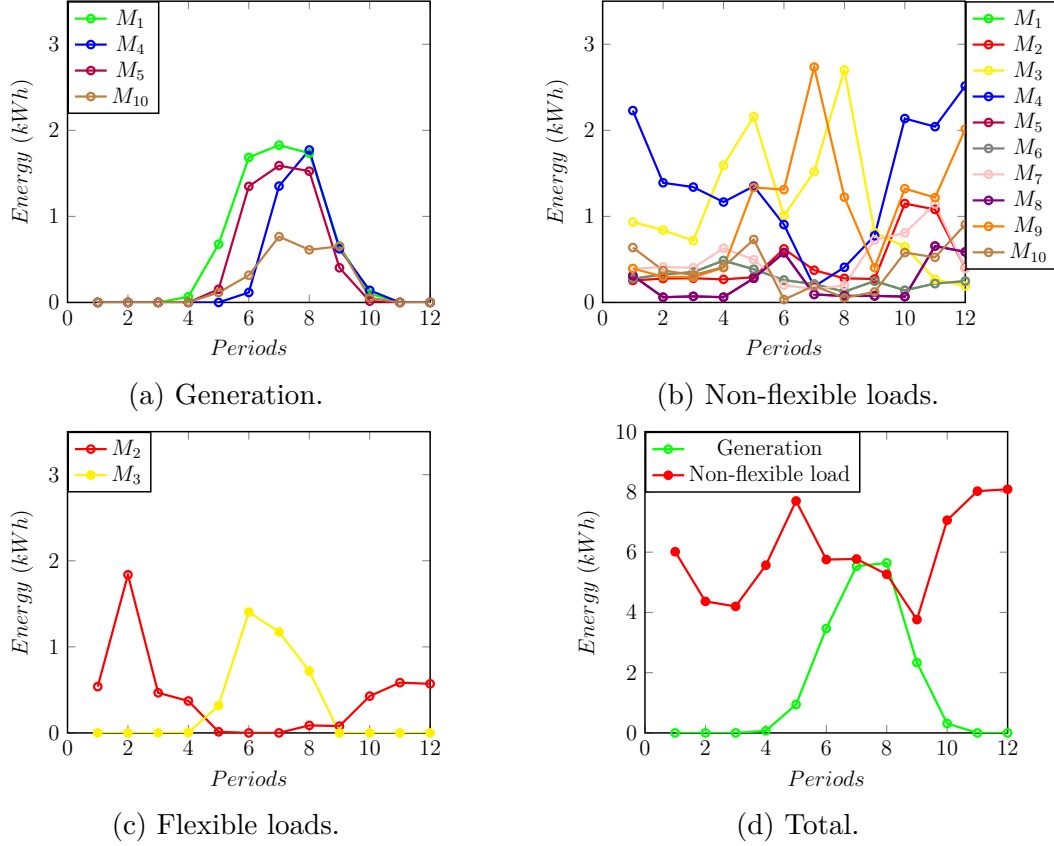


Figure 3: Data.

4.1.3 Setting

In this paper, the single-level problem (15) is solved using the Gurobi solver under the non-convex mode. We set the time limit to solve problem (15) to one hour, and our experiments are done on a processor Intel Xeon E312 (Sandy Bridge) CPU2.29GHz; each computation is paralyzed on ten threads. Section 4.2 presents the results.

4.2 Results

This section presents the results for the leader and the community members. We compare different scenarios, including the increase in generation and the presence of storage units in the community.

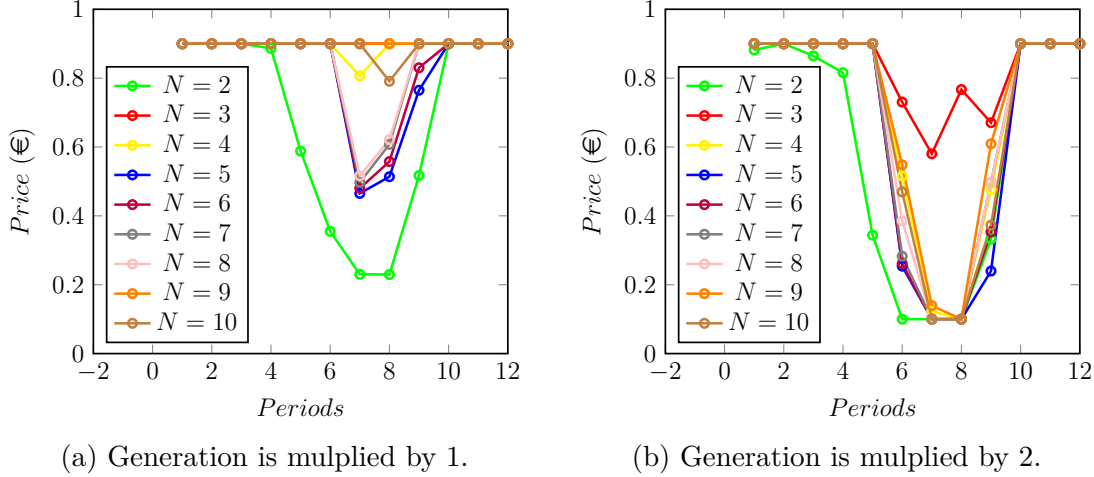
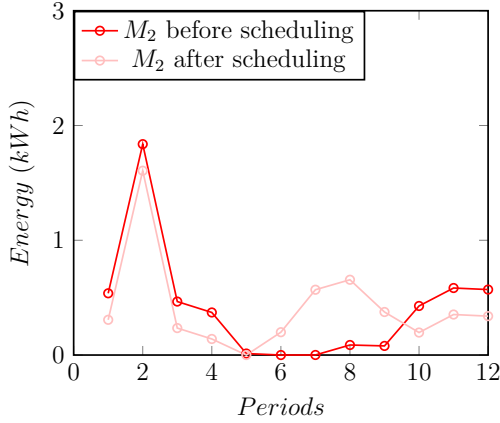


Figure 4: Leader's prices according to the community's generation.

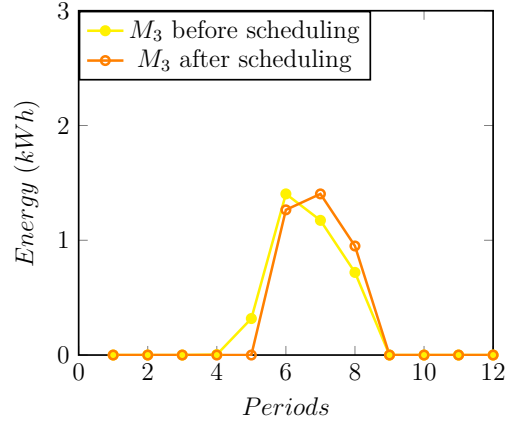
Table 2 and Figure 4, we present the leader's cost and prices for the grand coalition (with the coalition size N varying between 0 and 10) when the local generation is multiplied by 1 and 2. In Table 2, line $N = 0$ corresponds to the case where the leader does not interact with the community. The columns "CPU" and "Gap" display the CPU time required to find the leader's value while operating with the grand coalition and the gap to optimality. In addition, columns "Leader's value" and " $\Psi(N)$ " depict the leader and the grand coalition values. We report the values multiplied by -1 to simplify the tables. Therefore, a negative value in the tables corresponds to revenue, and a positive value represents an expenditure.

According to Table 2, it may not be profitable for the leader to operate with the community. This is because our instances are built such that the leader's periodic generation cost is $v_h^{\text{lead}} = 0.91$ for $h \in \mathcal{H}$ is greater than the upper bound on his interaction prices with the community $\bar{v}_h^s = \bar{v}_h^b = 0.9$ for $h \in \mathcal{H}$. Interacting with a community does not deteriorate the leader's value if $v_h^{\text{lead}} \leq \max\{\bar{v}_h^s, \bar{v}_h^b\}$ for $h \in \mathcal{H}$. Additionally, the energy collected by the leader from the community can be used to satisfy the leader demand, but this satisfaction is not considered in this paper. Further, we notice that increasing the generation improves both leader and community values. Finally, Figure 5 illustrates the demand response in the community.

Regarding cost allocation, as per Proposition 4, if $c_{nmh} = c_{mnh}$ for $n \in \mathcal{N}, m \in \mathcal{P}, h \in$



(a) Member 2 's flexible loads



(b) Member 3 's flexible loads

Figure 5: Demand response illustration if $N = 10$.

$\mathcal{H}$, each member $n \in \mathcal{N}$ receives $\tilde{\Pi}_n^F$ for each mechanism. Table 3 numerically illustrates this result for a community of ten members ($N = 10$), where each row corresponds to a community member. Columns “ $\tilde{\Pi}^F$ ” represent the members’ values defined in Definition 3.

Generation multiplied by 1					Generation multiplied by 2				
Size	Leader's value	$\Psi(N)$	CPU	Gap%	Size	Leader's value	$\Psi(N)$	CPU	Gap%
$N = 0$	54.06				$N = 0$	54.06			
$N = 1$	49.51	0.02	0.02	0	$N = 1$	44.18	-0.7	0.04	0
$N = 2$	52.17	5.83	t_l	0	$N = 2$	47.50	5.12	13.30	0
$N = 3$	54.08	17.70	9	0	$N = 3$	53.79	12.04	t_l	0
$N = 4$	54.07	28.32	34.40	0	$N = 4$	50.53	23.39	356	0
$N = 5$	53.33	26.99	927	0	$N = 5$	44.12	23.39	21	0
$N = 6$	53.46	29.39	1005	0	$N = 6$	44.47	25.36	34	0
$N = 7$	53.57	34.26	1820	0	$N = 7$	45.38	29.59	75	0
$N = 8$	53.60	36.53	188	0	$N = 8$	45.84	31.41	126	0
$N = 9$	54.11	47.18	18	0	$N = 9$	49.69	38.80	34	0
$N = 10$	54.08	49.12	104	0	$N = 10$	46.93	41.48	344	0

Table 2: EEU's and the community's value according to the generation.

Having flexible loads and storage units in the community makes it difficult to solve the single-level problem (15). Consequently, we assume there are no flexible loads in the community to find an optimal solution while considering storage units.

When storage is involved, we notice that having a storage unit in the community increases social welfare but not necessarily individual values. Indeed, if $N = 4$, i.e., members 1 to 4

Cost allocation*											
Generation multiplied by 1						Generation multiplied by 2					
Members	$\tilde{\Pi}^F$	SV	$\mathcal{B}^\#$	PCA	EPM	Members	$\tilde{\Pi}^F$	SV	$\mathcal{B}^\#$	PCA	EPM
1	-4.87	0	0	0	0	1	-3.33	0	0	0	0
2	8.24	0	0	0	0	2	6.99	0	0	0	0
3	14.14	0	0	0	0	3	8.19	0	0	0	0
4	10.75	0	0	0	0	4	11.65	0	0	0	0
5	-2.21	0	0	0	0	5	-0.62	0	0	0	0
6	2.54	0	0	0	0	6	2.15	0	0	0	0
7	5.00	0	0	0	0	7	4.36	0	0	0	0
8	2.32	0	0	0	0	8	1.99	0	0	0	0
9	11.26	0	0	0	0	9	7.52	0	0	0	0
10	1.95	0	0	0	0	10	2.58	0	0	0	0

*We present member's utility increase compared to $\tilde{\Pi}^F$ in percentage.

Table 3: Cost allocation if there are no storage units.

compose the community, $\Psi(\mathcal{N}) = 21.44$ without storage units, and $\Psi(\mathcal{N}) = 20$ if member 3 owns a BSS. However, while members 1 and 4's value increases with storage, members 2 and 4's value deteriorates. The leader's value also deteriorates when at least a member owns a competitive storage unit as shown in Table 4. Table 5 presents the cost allocation if there is a storage unit in the community according to the generation level.

Further, the grand coalition's value does not depend on the BSS owner, as Figure 6 shows. That is a consequence of Proposition 4. Indeed, as the leader maximizes his value, his interaction prices $(v_h^{\text{com}} = \tilde{v}_h^{\text{com}})_{h \in \mathcal{H}}$ which remain the same whether c is symmetric or not will discourage the storage for ulterior usage in the community; no BSS owner has the incentive to store energy and exchange it with other members. The phenomenon is presented in Figure 4a. Indeed, since the margin cost $C_3^{\text{BSS}} = 0.2$, charging and discharging a unit of energy leads to a cost at least equal to 0.4. By Setting the prices higher than 0.4, the leader ensures the BSS owner's profile has no impact on the social welfare.

For the four-member community, the grand coalition value is 21.88; it increases by about 2.1% by introducing a storage unit no matter who owns it.

5 Conclusion

This paper introduces a bi-level problem for energy communities interacting with an energy end-user; EEU. The energy end-user, whose pricing problem is the upper-level problem, must

Generation multiplied by 1 without storage					Generation multiplied by 1 with storage				
Size	Leader's value	$\Psi(N)$	CPU	Gap%	Size	Leader's value	$\Psi(N)$	CPU	Gap%
$N = 0$	54.06				$N = 0$	54.06			
$N = 1$	49.51	0.02	0.02	0	$N = 1$	49.51	0.02	0.01	0
$N = 2$	50.81	2.63	0.1	0	$N = 2$	50.81	2.63	0.09	0
$N = 3$	53.91	10.18	12	0	$N = 3$	53.32	10.18	t_l	3.3
$N = 4$	53.12	21.88	0.6	0	$N = 4$	53.32	21.43	t_l	0
$N = 5$	50.65	22.44	3	0	$N = 5$	51.78	20.87	t_l	0
$N = 6$	51.02	24.44	58	0	$N = 6$	52.01	22.99	t_l	7.1
$N = 7$	51.34	29.16	16	0	$N = 7$	52.19	27.70	3550	0
$N = 8$	51.50	31.35	8	0	$N = 8$	57.27	30.05	t_l	1.7
$N = 9$	53.92	39.71	10	0	$N = 9$	53.92	39.70	t_l	0
$N = 10$	53.43	42.30	167	0	$N = 10$	53.50	42.11	t_l	0
Generation multiplied by 2 with storage									
Size	Leader's value	$\Psi(N)$	CPU	Gap%					
$N = 0$	54.06								
$N = 1$	44.18	-0.7	0.09	0					
$N = 2$	45.69	1.97	0.33	0					
$N = 3$	51.78	6.51	t_l	9					
$N = 4$	49.21	16.74	t_l	8					
$N = 5$	43.57	16.01	t_l	6					
$N = 6$	44.51	18.91	t_l	9					
$N = 7$	44.82	21.32	t_l	16					
$N = 8$	45.22	25.29	t_l	13					
$N = 9$	48.37	32.16	t_l	13					
$N = 10$	46	34.2	t_l	7					

Table 4: EEU's and the community's value with and without storage units in the community.

collect the community's surpluses at time-varying prices to maximize revenue. We assume a centralized market design for the community, where a community coordinator aims to maximize the community's social welfare; the coordinator's problem is the lower-level problem. However, since the leader's prices are not differentiated per member, these prices depend on the electricity market's prices and the community's surpluses over time. Consequently, our model needs some fairness criteria that cannot be considered endogenously in the bilevel optimization problem. Instead, we model and analyze a cooperative game for cost allocation among the community members to ensure fairness. We study some of the game's properties and show that grand coalition stability can be assured for the Shapley value and variants and under mild assumptions for the other cost allocation mechanisms considered. Then, we

Cost allocation*											
Generation multiplied by 1						Generation multiplied by 2					
Members	$\tilde{\Pi}^F$	SV	$\mathcal{B}^\#$	PCA	EPM	Members	$\tilde{\Pi}^F$	SV	$\mathcal{B}^\#$	PCA	EPM
1	-3.5	0	0	0	0	1	-4.57	0	0	0	0
2	3.64	0	0	0	0	2	3.3	0	0	0	0
3	9.49	0	0	0	0	3	7.31	0	0	0	0
4	11.77	0	0	0	0	4	10.69	0	0	0	0
5	-1.02	0	0	0	0	5	-1.70	0	0	0	0
6	2.45	0	0	0	0	6	2.23	0	0	0	0
7	4.89	0	0	0	0	7	4.46	0	0	0	0
8	2.28	0	0	0	0	8	2.00	0	0	0	0
9	9.69	0	0	0	0	9	8.45	0	0	0	0
10	2.42	0	0	0	0	10	2.03	0	0	0	0

*We present member's utility increase compared to $\tilde{\Pi}^F$ in percentage.

Table 5: Cost allocation if there is a storage unit.

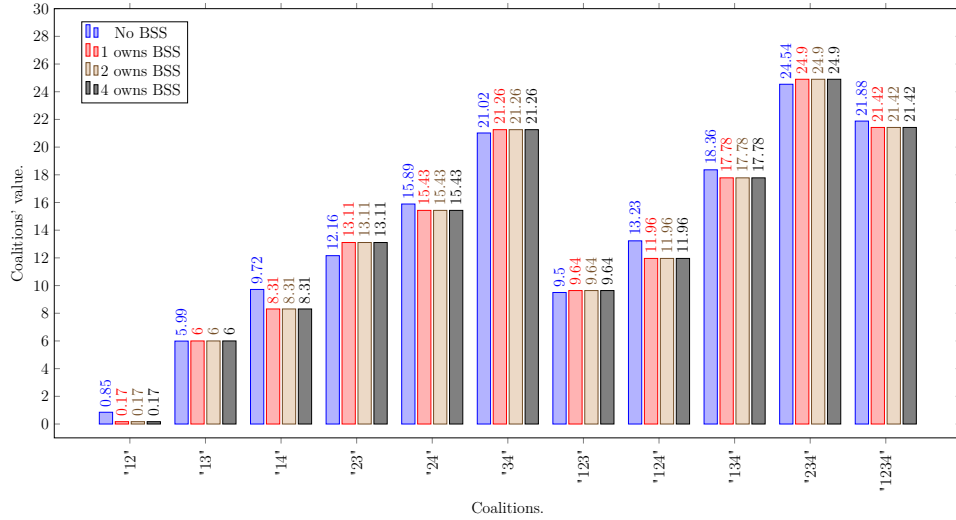


Figure 6: Coalition's values according to the BSS owner for a four-member community.

study some allocation mechanisms' properties and allocate the community's value according to the mechanisms belonging to the core. Finally, we illustrate our results with some numerical experiments on real-world instances that show the benefits for both the leader and the community of participating in such a local system.

This paper assumes that the leader could take 100% of the community surplus. However, in practice, grid operators are more concerned with peak reduction. It would be valuable to investigate the effects of limiting peak loads and community injections from a broader

perspective. Additionally, rather than treating the parameters as deterministic, an alternative approach would be to consider uncertainty in generation, mainly since communities can produce renewable energy and uncertainty in demand.

Declaration

Conflict of interest: The authors declare that they have no financial interests or personal relationships that could have influenced the work presented in this article.

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